

GAIN-DANN: A Domain-Adversarial Generative Model for Missing Data Imputation in Proteomics

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Missing values are one of the most pervasive challenges in proteomics, hiding biological information and reducing the reliability of downstream analysis. Traditional imputation methods, such as missForest, become computational demanding as the size of the dataset grows, making them unfeasible for large-scale proteomics datasets. Generative Adversarial Imputation Networks (GAIN) have emerged as a promising alternative to this problem. Previous work in our group extended GAIN to proteomics data, ProtoGain, and demonstrated promising results, laying the foundation for the present study. This work proposes GAIN-DANN, a hybrid model that integrates GAIN with Domain-Adversarial Neural Networks (DANN) to improve missing values imputation in proteomics.

The model was designed to exploit shared information across related domains and datasets, aiming to learn domain-invariant representations capable of generalizing to new datasets. In the intra-dataset evaluation, the GAIN-DANN achieved lower reconstruction error than the baseline ProtoGain while simultaneously mitigating batch effects, suggesting that the adversarial nature of the model enabled it to focus on biological rather than domain-specific variation. In the cross-dataset setting, where pretrained representations were reused across datasets, the GAIN-DANN reduced the computational time by approximately 75% and achieved a substantial lower RMSE than ProtoGain models trained from scratch. These results demonstrated that previously learned representations can be successfully reused to improve imputation in smaller datasets. Overall, the results highlight the potential of combining adversarial domain adaptation with generative imputation to enhance missing values imputation.

Additional Key Words and Phrases: Generative Adversarial Imputation Network, Domain-Adversarial Neural Network, Domain Adaption, Missing Values Imputation

Since the sequencing of the first protein by Frederick Sanger in the 1950s, proteomics has evolved into a cornerstone of biological research, helping discoveries in areas such as drug development and biomarker discovery. Advances in mass spectrometry have enabled large-scale proteomics studies, however these datasets usually suffer from missing values. Such missingness can arise from multiple sources including instrument sensitivity, detection limits, and/or differences in experimental setups. This represents one of the most pervasive challenges in proteomics, often obscuring biological information and compromising downstream analysis.

Traditional imputation methods, such as missForest, have been widely applied but become computationally demanding as the size of the datasets increases. Generative Adversarial Imputation Networks have emerged as promising alternative to this problem. ProtoGain¹ successfully extended GAIN to proteomics data. However, training these models from scratch for every new dataset can be inefficient, especially in smaller and noisier datasets, which are more prone to overfitting. This work explores a strategy to enhance imputation performance by enabling the model to capture shared information across related domains, thereby improving imputation performance.

¹<https://github.com/QuantitativeBiology/GenerativeProteomics>

1 Related work

Generative Adversarial Imputation Networks extend the Generative Adversarial Network (GAN) framework to handle missing data [5]. GAIN consists of a generator and discriminator trained adversarially to impute missing values. The generator receives partially observed data, a binary mask, and random noise, producing imputed values that the discriminator attempts to distinguish from observed entries using a hint mechanism. Through this minimax game, the generator learns to produce realistic imputations consistent with the observed data.

ProtoGain, developed by our group, adapted GAIN to real proteomics data [4], achieving competitive root mean square error (RMSE) with substantially lower computational cost than traditional imputation methods such as missForest.

Domain-Adversarial Neural Networks, proposed by Ganin et al., address domain shifts by embedding domain adaptation into the representation learning directly [3]. A gradient reversal layer (GRL) ensures that the encoder learns representations that are discriminative for the main task and while being invariant to the domain shifts. This allows knowledge transfer between domains with different distributions. This principle is one of the foundations of the proposed GAIN-DANN model, helping to improve imputation quality and mitigate batch effects across proteomics domains.

2 The GAIN-DANN Model

Proteomics datasets are often characterized by substantial missingness and domain-specific variability introduced by differences in the experimental design, instrumentation and sample preparation. While ProtoGain has demonstrated strong performance when trained and applied within individual datasets, we hypothesized that its effectiveness can be further enhanced by leveraging information learned from other related proteomics datasets. In particular, knowledge learned from larger and more complete datasets can improve imputation in smaller and/or noisier datasets. However, effectively transferring such knowledge is non-trivial due to distributional shifts and differences in dimensionality between datasets. To address this challenge, we propose a hybrid model, GAIN-DANN, that integrates the representation learning capabilities of the Domain-Adversarial Neural Networks with the imputation mechanism of GAIN. The proposed model is designed to promote domain-invariant feature learning while maintaining an accurate reconstruction of the missing protein intensities, thereby enabling cross-project transfer for improved imputation.

The intuition behind the proposed model stems from the premise that imputation quality can be improved by transferring relevant knowledge across proteomics datasets. Standard imputation models such as GAIN are typically trained independently on each dataset, learning local distributions. By leveraging knowledge from other related datasets, it becomes possible to capture information that may be missing or underrepresented in a single dataset, thus producing

imputations that are more consistent with true biological variation. The DANN framework was developed to mitigate domain shifts by encouraging the emergence of latent representations that are both discriminative for the main learning task and invariant with respect to the domain. By considering this principle into the imputation process, our model leverages shared biological information across related datasets while minimizing dataset-specific information. This design allows the model to benefit from representations learned on larger or more complete datasets, ultimately enhancing imputation quality in small, highly incomplete target domains.

The canonical DANN model was originally designed for classification tasks, as such the task-specific module in the original DANN is a label predictor. In our adaptation, we replaced the label predictor by a GAIN module, shifting the focus from classification to missing values imputation. This allows the model to learn latent representations that remain aligned across domains while learning to impute the missing values.

2.1 Model Design

This section describes the design of the proposed model, detailing its main modules and overall structure. The design was guided by the need to effectively address missing data in proteomics while accounting for domain shifts, resulting in an architecture that combines generative imputation and adversarial domain adaptation mechanisms. Section 2.1.1 details both the input representation and the preprocessing steps to prepare the data for model training. Section 2.1.2 presents the model architecture.

2.1.1 Input. The framework processes data differently for training and imputation, reflecting the distinct requirements of each task. The input consists of a protein abundance matrix in a CSV format, with samples as rows and proteins as columns. For training, an additional domain label column ("Domain") is required for the domain-adversarial component of the model. For imputation, only the protein abundance matrix is needed. Although this is not a model requirement, log-transformed data is recommended over non-transformed data. The data is standardized using a Standard Scaler during both the training and imputation stages. Finally, during training, artificial missingness is induced (default 10%) to have ground-truth values to compare the predictions.

2.1.2 Architecture. The GAIN-DANN architecture was designed with two primary objectives: (i) to accurately impute missing protein abundances, and (ii) to ensure that the imputations are invariant to domain shifts. The model integrates five main components: an encoder, a gradient reversal layer, a domain classifier, a GAIN module, and a decoder. Each of these modules contributes to a complementary aspect of the imputation and domain adaptation process.

The encoder maps protein abundance vectors, $\mathbf{x} \in \mathbb{R}^m$, into a latent representation $\mathbf{z} \in \mathbb{R}^D$, where $D = m$, that learns domain-invariant representations of the data. Its design combines fully connected layers with batch normalization, ReLU activation, and dropout regularization to enhance generalization and stability.

The domain classifier is a multi-class logistic regressor, $G_y(\cdot; \theta_y)$, that aims to distinguish between the domains of the samples. It does not receive the latent space vector $\mathbf{z}$ directly, but receives it

indirectly through the Gradient Reversal Layer. This step is crucial since during training the GRL inverts the gradients from the classifier, forcing the encoder to learn domain-invariant features that maximize the domain confusion of the domain classifier while retaining discriminative power for imputation.

The GAIN module, adapted from ProtoGain, performs the imputation within the latent space. It uses an adversarial mechanism between a generator and a discriminator to infer missing entries, while leveraging mask and hint matrices to improve accuracy.

The imputed latent representations, $\hat{\mathbf{z}}$, are subsequently passed through the decoder, which reconstructs the complete protein abundance vector $\hat{\mathbf{x}} \in \mathbb{R}^m$, ensuring compatibility with downstream proteomics analysis. The decoder mirrors the structure of the encoder.

2.2 Training

The training of GAIN-DANN simultaneously optimizes two objectives: (i) learning domain-invariant representations to enable knowledge transfer, and (ii) accurately imputing missing protein abundances. These objectives are guided by distinct loss terms computed across both the latent and original feature spaces.

To correctly handle missingness, two masks are used: the true missingness mask M_{true} , indicating genuinely missing entries in the original dataset, and the induced missingness mask M_{ind} , defining artificially removed values for training. While M_{true} is used to assess reconstruction in the protein space, M_{ind} supervises imputation in the latent space.

To enforce domain invariance, the gradient from the domain classifier is reversed during backpropagation and scaled by a factor $-\lambda_p$, ensuring adversarial learning between encoder and classifier. The domain adaptation coefficient λ_p follows the DANN schedule described in Equation 1, where p represents the training progress linearly increasing from 0 to 1.

$$\lambda_p = \frac{2}{1 + \exp(-10p)} - 1 \quad (1)$$

The GAIN module imputes missing values in latent space, producing $\hat{\mathbf{z}} = G_{\text{GAIN}}(\mathbf{z}; \theta_f)$. It is trained using the induced mask M_{ind} , minimizing the imputation loss (Equation 2).

$$\mathcal{L}_{\text{imputation}}(\theta_f) = \frac{\sum_{i=1}^m \|M_{\text{ind}}^{(i)} \odot (\hat{\mathbf{z}}_i - \mathbf{z}_i)\|^2}{\sum_{i=1}^m M_{\text{ind}}^{(i)}} \quad (2)$$

The decoder then reconstructs the protein space as $\hat{\mathbf{x}} = G_d(\hat{\mathbf{z}}; \theta_d)$, evaluated by the reconstruction loss described in Equation 3, where MSE corresponds to the mean squared error.

$$\begin{aligned} \mathcal{L}_{\text{reconstruction}}(\theta_d, \theta_f) &= \sqrt{\frac{\sum_{i=1}^m \|M_{\text{true}}^{(i)} \odot (\hat{\mathbf{x}}_i - \mathbf{x}_{\text{true}_i})\|^2}{\sum_{i=1}^m M_{\text{true}}^{(i)}}} \\ &= \sqrt{\text{MSE}(\hat{\mathbf{x}} \odot M_{\text{true}}, \mathbf{x}_{\text{true}} \odot M_{\text{true}})} \end{aligned} \quad (3)$$

The overall model objective is a weighted minimax problem between the imputation, reconstruction and domain losses (Equation 4). To prioritize imputation over domain adaptation the constraint $\alpha, \beta > \gamma$ was imposed.

$$\min_{\theta_f, \theta_d} \max_{\theta_y} \mathcal{L}_{\text{model}} = \alpha * \mathcal{L}_{\text{imputation}} + \beta * \mathcal{L}_{\text{reconstruction}} + \gamma * \mathcal{L}_{\text{domain}} \quad (4)$$

The training procedure is detailed in the pseudo-code provided in Algorithm 1 and the overall architecture of the proposed framework is illustrated in Figure 1. The GAIN component follows the generic training procedure of GAIN, where the discriminator and generator are updated through the corresponding update functions implemented in the ProtoGAIN module.

Training is implemented in PyTorch using an AdamW optimization with gradient clipping, early stopping, and a ReduceLROnPlateau scheduler based on validation RMSE to dynamically adjust learning rate and prevent overfitting.

Algorithm 1: Stochastic gradient descent training procedure of GAIN-DANN

Input: Protein abundance matrix $X \in \mathbb{R}^{n \times m}$;
Protein abundance matrix with induced missingness
 $X_{\text{ind}} \in \mathbb{R}^{n \times m}$;
Domain labels $Y \in \mathbb{R}^n$;
Learning rate η

```

1 while stopping criterion not met do
2   for  $i \leftarrow 1$  to  $n$  do
3      $\mathbf{x}_i \leftarrow X_{\text{ind}}[i]$ ,  $\mathbf{x}_{\text{true}} \leftarrow X[i]$ 
4      $y_i \leftarrow Y[i]$ 
5      $M_{\text{true}}[i, j] \leftarrow 1$  if  $X[i, j]$  observed, else 0
6      $M_{\text{ind}}[i, j] \leftarrow 1$  if  $X_{\text{ind}}[i, j]$  observed, else 0
7     # Forward Propagation
8      $\mathbf{z}_i \leftarrow G_f(\mathbf{x}_i; \theta_f)$  ▷ Encoder ( $G_f$ )
9      $\hat{y}_i \leftarrow G_y(\mathcal{R}(\mathbf{z}_i); \theta_y)$  ▷ Domain Classifier ( $G_y$ )
10    # Update of GAIN
11     $\hat{\mathbf{z}}_i \leftarrow G_{\text{GAIN}}(\mathbf{z}_i; \theta_f)$  ▷ GAIN ( $G_{\text{GAIN}}$ )
12     $\hat{\mathbf{x}}_i \leftarrow G_d(\hat{\mathbf{z}}_i; \theta_d)$  ▷ Decoder ( $G_d$ )
13    # Backward Propagation
14    # Compute Losses
15     $\mathcal{L}_{\text{domain}}(\theta_y, \theta_f) \leftarrow \text{CrossEntropy}(\hat{y}_i, y_i)$ 
16     $\mathcal{L}_{\text{imputation}}(\theta_f) \leftarrow \text{MSE}(\hat{\mathbf{z}}_i \odot M_{\text{ind}}^{(i)}, \mathbf{z}_i \odot M_{\text{ind}}^{(i)})$ 
17     $\mathcal{L}_{\text{reconstruction}}(\theta_d, \theta_f) \leftarrow$   

     $\text{RMSE}(\hat{\mathbf{x}}_i \odot M_{\text{true}}^{(i)}, \mathbf{x}_{\text{true}} \odot M_{\text{true}}^{(i)})$ 
18     $\mathcal{L}_{\text{model}}(\theta_f, \theta_y, \theta_d) \leftarrow$   

 $\alpha * \mathcal{L}_{\text{imputation}} + \beta * \mathcal{L}_{\text{reconstruction}} + \gamma * \mathcal{L}_{\text{domain}}$ 
19    # Update parameters
20     $\theta_d \leftarrow \theta_d - \eta \nabla_{\theta_d} \mathcal{L}_{\text{reconstruction}}$ 
21     $\theta_y \leftarrow \theta_y - \eta \lambda_p \nabla_{\theta_y} \mathcal{L}_{\text{domain}}$ 
22     $\theta_f \leftarrow \theta_f - \eta (\nabla_{\theta_f} (\alpha * \mathcal{L}_{\text{imputation}} + \beta * \mathcal{L}_{\text{reconstruction}})$   

 $\quad - \lambda_p \nabla_{\theta_f} \mathcal{L}_{\text{domain}})$  ▷  $-\lambda_p$ , since the  

    GRL inverts the gradient
23  end
24 end
```

2.3 Evaluation

We employed a five-fold stratified cross-validation procedure. During training, several metrics were monitored at each epoch: the imputation loss ($\mathcal{L}_{\text{imputation}}$), domain loss ($\mathcal{L}_{\text{domain}}$), domain classification accuracy, total model loss ($\mathcal{L}_{\text{model}}$), task-specific loss ($\mathcal{L}_{\text{task-specific}}$), and reconstruction loss ($\mathcal{L}_{\text{reconstruction}}$, measured as RMSE). A decreasing $\mathcal{L}_{\text{imputation}}$ and $\mathcal{L}_{\text{reconstruction}}$ indicate improved imputation and reconstruction, while an increasing $\mathcal{L}_{\text{domain}}$ with classification accuracy approaching random chance ($\frac{1}{L}$ for L domains) reflects successful domain invariance. Oscillations in $\mathcal{L}_{\text{model}}$ are expected due to the adversarial training dynamics.

Model performance was primarily assessed through RMSE and the Pearson correlation coefficient computed between observed and imputed protein abundances on test samples. Predictions were inverse-transformed to the original log-scale prior to evaluation. Metrics were aggregated by domain within each fold and averaged across folds to yield a more reliable estimate of the generalization performance the model.

Finally, potential batch effects were examined using UMAP visualizations of the latent representations before and after training. The disappearance of domain-driven clustering after training suggests that the encoder successfully learned domain-invariant representations.

2.4 Inference

Once the model has been trained, it can be used to perform imputation on unseen proteomics datasets. The input is a protein abundance matrix formatted as a CSV file where rows correspond to samples and columns correspond to proteins, consistent with the training input format. Unlike training, no "Domain" column is required, since domain labels are only necessary to enforce the adversarial objective during training. The inference data should be processed using the same transformation as the training data to ensure consistency.

During inference, the domain-adversarial components (domain classifier and GRL) are disabled, leaving only the modules responsible for imputation active (encoder, GAIN and the decoder). As such, the model focuses exclusively in reconstructing the missing protein intensities, while the representations remain domain-invariant as learned during training.

The resulting matrix $\hat{X}$ contains both the observed entries and the imputed values for the initial missing entries. Since the model operates on normalized data, the inverse transformation of the scaler used during training must be applied to return the data to its original protein space. Finally, the imputed dataset is stored as a CSV file with the samples as rows and proteins as columns under a Result directory.

2.5 Hyperparameters Optimization

Hyperparameter optimization was performed using Optuna [1] for 50 trials, with a two-fold cross-validation scheme and the average best validation RMSE across folds as the objective metric. To reduce computational cost, the optimization was conducted on a smaller subset of the HeLa dataset, described in Section 3.1.1, using only the last 1,000 proteins. However, it is important to note that by using

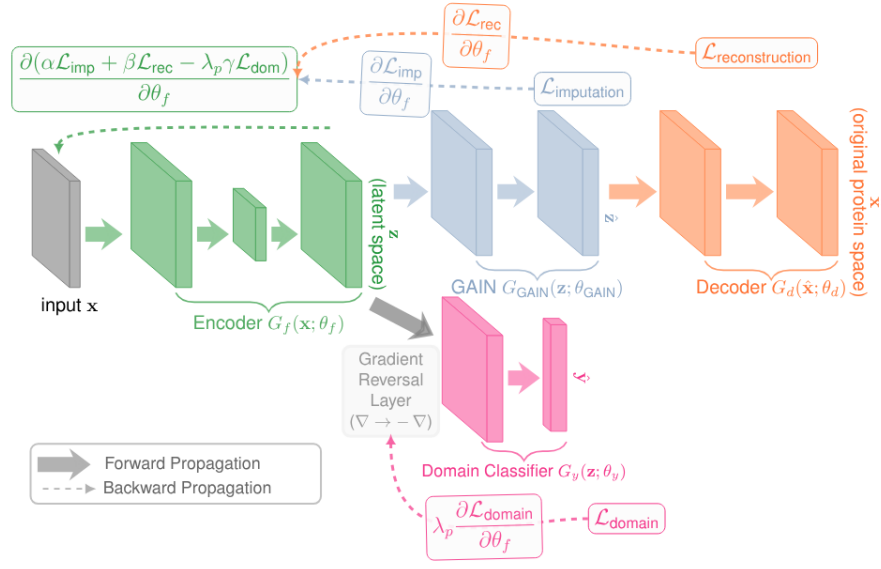


Fig. 1. The proposed GAIN-DANN architecture includes an encoder (green), a gradient reversal layer (gray) inserted between the encoder and domain classifier (pink), a GAIN module (blue), and a decoder (orange). For clarity, loss terms are abbreviated as $\mathcal{L}_{rec}$ for the reconstruction loss, $\mathcal{L}_{imp}$ for the imputation loss, and $\mathcal{L}_{dom}$ for the domain loss.

a reduced subset it may introduce some degree of bias. To further limit the computational time, the maximum number of epochs was limited to 100. The optimal values were: 16 for batch size, 3.16×10^{-5} for the learning rate, 0.03 for the weight decay, two hidden layers, a dropout rate of 0.3, $\alpha = 3$, $\beta = 3$ and $\gamma = 1$. The optimization revealed that the GAIN loss and reconstruct loss should contribute equally to the model loss. However, it is important to note that even though it is ideally to perform the hyperparameter prior to experiments, it is both a computationally and time intensive operation. As such, the hyperparameter optimization was the last analysis performed. Throughout the experiments, the hyperparameters were manually adjusted, the values used in each experiment are specified. Notably, the manually selected values were relatively close to the optimal values.

2.6 Imputation with a pretrained model

It is well understood that proteins are biological entities shared across experiments and conditions, in this work, we hypothesized that models trained on one dataset can potentially capture transferable protein-level representations to use on other datasets, thereby improving the imputation process.

To this end, we consider the source dataset as the data on which the model is initially trained, and the target dataset refers to the dataset to be imputed. In all experiments, the pretraining and fine-tuning were conducted using our proposed GAIN-DANN model.

We evaluated three fine-tuning strategies, each of increasing complexity, to transfer the pretrained knowledge to the target dataset. For all strategies, the imputations were restricted to the intersection of proteins present in both the source and the target dataset. To preserve the dimensionality expected by the trained model and

avoid discarding learned representations, the target dataset was subsequently padded to match the source dimensionality, with the proteins not present in the target dataset represented as NaN.

2.6.1 Full Encoder Freezing. In an initial stage, we froze the encoder entirely and retrained only the decoder and the GAIN module on the target dataset. This approach assumes that the pretrained encoder already captures a sufficiently general representation of the protein features.

2.6.2 Partial Encoder Freezing. The second strategy unfreezes the last two layers of the encoder. This is motivated by the learning patterns of feedforward neural networks, where the initial layers tend to capture more general patterns whereas the last layers capture more dataset-specific characteristics. By fine-tuning the last two layers, the model can adapt its representations to the target dataset without discarding the general knowledge obtained from the source dataset. Since more parameters are now updated, this strategy was trained for a higher number of epochs compared to the Full Encoder Freezing strategy.

2.6.3 Domain-Adversarial Adaptation. The third strategy fully exploits the adversarial architecture of our model. Instead of only retraining on the target dataset, we provide both the source and target datasets to the model during fine-tuning. That way, the model is retrained with the objective of minimizing the discrepancies between the source and target datasets, allowing the learned representation to be adapted taking into consideration the target distribution.

In order to accommodate this setup, the model must distinguish among a larger number of domains. Consequently, the domain classifier module is reinitialized, with the number of output classes

updated to *total number of domains = number of source domains + number of target domains*. To avoid hindering the encoder’s learning process, the hidden dimension of the domain classifier was set to 64. Apart from reinitializing the domain classifier, the model architecture remained the same as the pretrained architecture. During fine-tuning, the parameters of all trainable modules were updated to adapt to the target dataset.

2.7 Implementation and Usage Details

Although the core contribution of this work is methodological, we also provide an implementation designed to facilitate practical usage, in particular for imputation with a pre-trained model. The full implementation and detailed execution instructions are available in the public repository². Furthermore, as part of a collaborative effort, a thesis developed in parallel within the same project has made a pre-trained version of the model publicly available on the Hugging Face platform³ to facilitate its use and testing by the community. The implementation is organized around two components: JSON configuration files and command-line execution modes. The configuration files define dataset related parameters and training or inference hyperparameters (e.g. learning rate, dropout probability), and parameters (e.g. number of epochs, number of folds). The framework supports four execution modes: `train` and `save` allowing users to train and save new models, `impute` to predict missing values using a trained model, and `finetune` allowing a pre-trained model to be adapted to a new dataset through a specified fine-tuning strategy.

3 Results

This section presents the results obtained with the proposed model and discusses its performance in proteomics data imputation. Section 3.1 provides a detailed description and analysis of the datasets used throughout the experiments. Section 3.2 presents the results and analysis of both intra-dataset and cross-dataset experiments. Finally, Section 4 discusses the benefits and limitations of the proposed model.

3.1 Datasets

In the following sections, we describe and analyze the datasets used for the development of our model 3.1.1 and the datasets used to evaluate imputation with a pretrained model 3.1.2. All the datasets were provided by PRIDE⁴ at EMBL-EBI.

3.1.1 HeLa dataset. A HeLa dataset was provided with 4,820 samples across 11,013 proteins in an `anndata` format with multiple transformations associated with the data. We chose the log-transformed (*IbaqLog*) layer, since the raw intensity values are spread over several orders of ($[0, 10^{10}]$), and log-transformed data is a common choice in proteomics settings. Each of the sample is uniquely identified, following the format `<Project ID>-Sample-<Sample Number>`, linking it to the corresponding project. The dataset has 17 projects, that will be considered as domain for the proposed model.

The distribution of nonzero protein intensities follows approximately a normal distribution center around the log-intensity values of 5 and 7. The protein intensities distributions across projects are highly homogeneous, exhibiting a mean of 5.82 ± 0.74 (mean $\pm$ std). However, the overall distribution of values is not homogeneous throughout the whole range of values. Zeros cluster at the tail of the distribution and the total number of zeros is an order of magnitude higher than the rest. Given these observations and the inherent characteristics of mass spectrometry, where missing values often occur, we interpreted the zeros as missing values rather than as a true absence of expression. This resulted in an overall missingness of approximately 70%.

The dataset revealed extensive missingness: half of the samples (2,412) contained more than 71% missing protein intensities, with a strongly right-skewed distribution dominated by samples exhibiting 65–75% missingness. Only three samples showed less than 20% missingness. At the protein level, missingness followed a bimodal distribution, suggesting that the missingness is not random but likely related to detection sensitivity in mass spectrometry. This reflects the polarized nature of missingness in proteomics data: proteins are either well-detected or poorly detected. Over 60% of proteins (6,661) had missingness above 90%, corresponding to systematically lower intensities (mean 4.31) compared to low-missing proteins (mean 6.19). Missingness across projects ranged from 20% to 86%, showing no direct relation with sample size, this variation is likely explained by differences in the experimental setups.

The dataset is also highly imbalanced. Project PXD042233 alone accounts for 88% of all samples (4,258 out of 4,820), whereas the majority of the projects have less than 50 samples. Any imbalance representation of the domain classes, not exclusively in this dataset, can bias the learning process towards domain-specific characteristics rather than promoting domain-invariant representations. This was taken into account in the training process. In order to both prevent the bias and to preserve the domains proportions, a weighted sampler was employed during the data splitting. We further investigated the imbalance influence by conducting experiments with and without PXD042233 (see Section 3.2.1).

Since all samples derive from the same biological source (HeLa cells), substantial overlap of the samples across projects in a UMAP would be expected. Instead, the UMAP projection reveals a clear clustering by project rather than biological variation, revealing that the dataset suffers from batch effects. The presence of these batch effects provided further motivation for incorporating domain-adaptation strategies to mitigate project-specific biases and to promote learning of domain-invariant representations.

These observations provided motivation for employing domain-adaptation strategies with the adversarial training, to promote domain-invariant representations and more robust imputations across heterogeneous projects.

3.1.2 PRIDE datasets. We also had multiple PRIDE proteomics datasets of different tissues types available provided by the EMBL-EBI (Table 1). These datasets were used to evaluate the cross-domain scenarios described in Section 2.6. As shown in Table 1, the datasets have a wide range of tissue types. To assess the effect of transferring knowledge from a pretrained model to a target dataset of the same

²<https://github.com/QuantitativeBiology/GenerativeProteomics>

³https://huggingface.co/QuantitativeBiology/GAIN_DANN_model

⁴<https://www.ebi.ac.uk/pride/>

Project	Tissue	Number of Samples	Number of Proteins	Missing Rate (%)
PXD005736	Heart	2	8,470	2.10
PXD002179	Testis	3	8,836	3.38
PXD012131	Brain	26	10,188	6.77
PXD005445	Brain	7	9,165	7.42
PXD008840	Stomach	84	9,278	14.24
PXD010154	Multiple	38	14,669	17.15
PXD004452	Multiple	4	8,657	17.44
PXD019909	Skin	12	11,700	18.62
MSV0000858	Multiple	460	13,051	29.62
PXD006675	Heart	56	11,135	30.81
PXD020192	Multiple	46	6,177	34.88
PXD000612	HeLa-S3	11	8,782	39.62
PXD030304	Multiple	2,012	8,754	39.93
PXD000865	Multiple	34	8,680	60.51
PXD000561	Multiple	85	12,025	80.23

Table 1. PRIDE datasets, ordered by ascending missing rate. Datasets annotated as "Multiple" in the Tissue Type include samples from more than one tissue type. Tissue types in bold correspond to datasets containing a single tissue type.

tissue type, we filtered datasets corresponding to specific tissue types.

3.2 Evaluation

The proposed model was evaluated in two different settings:

- **Intra-Dataset** where the model was evaluated on the same dataset with multiple related domains. This assesses the model ability to learn domain-invariant representations and generalize across domains. The HeLa dataset was used since it has samples from the same biological source across multiple projects (domains).
- **Cross-Dataset** in which a pretrained model on a source dataset is used to help impute missing values in a target dataset. This assesses if pretrained representations can improve imputation performance.

All the experiments were conducted on a NVIDIA RTX A4000 GPU with 16 GB of RAM.

3.2.1 Intra-dataset. To assess the model ability to generalize across domains within the same biological source, we conducted an intra-dataset evaluation using two variants of the HeLa dataset: **HeLa-Reduced** and **HeLa-Reduced-NoPXD042233**.

Due to computational limitations, the HeLa-Reduced dataset was obtained by excluding the 2,000 proteins with the highest proportion of missing values, resulting in 4,820 samples and 9,013 proteins and an overall missingness of 63.43%. Since the project PXD042233 accounts for the majority of the samples, we repeated the same experiments after excluding it, to assess whether by having more samples it enforces more pressure on the model to distinguish it from the

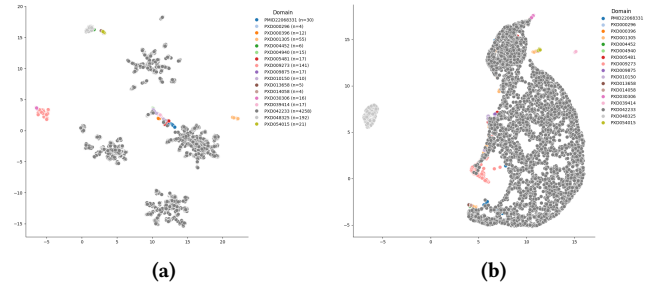


Fig. 2. UMAP projection (a) of the HeLa-reduced, (b) of the latent representation of the HeLa-Reduced after training of our model.

other projects, potentially influencing the learned latent representation. This new variant, referred to as HeLa-Reduced-NoPXD042233, comprises 562 samples across 9,013 proteins and an overall missingness of 58.65%.

A GAIN-DANN model was trained in each dataset for 200 epochs, with an early stopping patience of 10 epochs, with the following hyperparameters: learning rate of 1×10^{-4} , two hidden blocks for both the encoder and decoder, dropout rate of 0.3, $\alpha = 2.0$, $\beta = 3.0$ and $\gamma = 0.5$.

For evaluation, an additional 10% of the observed protein intensities were randomly masked in each dataset. This modified dataset was used to assess the performance of both the GAIN-DANN model and a baseline ProtoGain model. The baseline model was trained with all the default hyperparameters. All reported results correspond to the models performances on the datasets with the additional 10% induced missingness.

HeLa-Reduced

Since all samples are derived from the same biological source (HeLa cells), a substantial overlap among samples from different projects would be expected in a UMAP representation. Instead, the UMAP projection reveals a clear clustering by project revealing that the dataset suffers from batch effects (Fig. 2(a)). One of the implicit objectives of our model is to learn domain-invariant features through the encoder so that the learned representation can generalize across domains. Consequently, we expect that our model can mitigate the batch effects.

Indeed, the UMAP projection of the latent space, after the training of our model, shows that batch effects were almost completely mitigated (Fig. 2(b)). Two distinct clusters can be observed: the rightmost cluster contains samples from multiple domains, suggesting that it no longer reflects project identity, while the leftmost cluster consist exclusively of samples from PXD048325.

To understand why the project PXD048325 remains distinguishable, we compared the distribution of protein intensities between this project and the remaining ones. Project PXD048325 exhibits a mean intensity of 4.90 ± 1.04 (mean $\pm$ std), whereas the other projects have an average mean of 5.91 ± 0.64 . Interestingly, another project, PXD054015, is also quite distinct from the others with a mean of 4.86 ± 1.09 , despite clustering with the main cluster. Both projects present relatively low missingness rates, PXD054015 has 20% and PXD048325 has 22%, but differ significantly in the sample

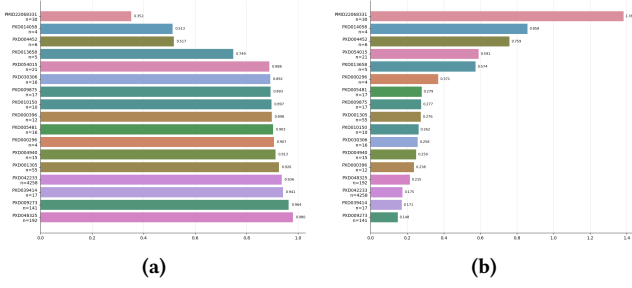


Fig. 3. **Imputation quality.** Results for each individual project on the HeLa-Reduced dataset after training the GAIN-DANN model with a two-fold cross-validation (a) Pearson correlation coefficient, (b) RMSE. Each bar represents one project, with the corresponding number of samples indicated below the project ID (n = number of samples).

size, with the former having only 21 samples versus 192 samples for the PXD048325.

To further investigate whether the PXD054015 could be influenced by PXD042233, the largest project, we quantified the similarity of their protein coverage patterns. For each project, a coverage vector was obtained by calculating for every protein the fraction of samples with non-missing abundance values. Let $C_i \in [0, 1]^m$ denote the coverage vector of project i , where each element represents the proportion of non-missing values for a given protein. The similarity between two projects, i and j , was then defined as the Pearson correlation coefficient between their coverage vectors (Equation 5).

$$\text{sim}(i, j) = \text{corr}(C_i, C_j) \quad (5)$$

The similarity coefficients $\text{sim}(\text{PXD054015}, \text{PXD042233}) = 0.429$ and the $\text{sim}(\text{PXD048325}, \text{PXD042233}) = 0.456$ indicate that both projects exhibit a similar coverage with respect to PXD042233. Thus the coverage similarity alone does not explain the clustering pattern. A more plausible explanation is that the limited number of samples in PXD054015 constrains its representation in the latent space, and that with more samples, this project might have diverged further from PXD042233, resembling the observed pattern for PXD048325.

As for the imputation quality, we analyze the average Pearson correlation and RMSE across folds for each individual project (Figures 3(a) and 3(b)). As anticipated, projects with lower Pearson coefficients generally coincided with higher RMSE values. The Pearson correlation quantifies the linear association between predicted and observed protein intensities in the original protein space, whereas the RMSE captures the reconstruction error in the standardized space. Jointly, these metrics provide a more comprehensive view of the model performance.

A clear trend emerges with respect to the sample size. Larger projects, such as PXD048325 ($n = 192$) and PXD009273 ($n = 141$) achieved the best overall performance, with Pearson correlations above 0.96 and RMSE values below 0.22, indicating that the imputations are highly accurate. In contrast, smaller projects, particularly those with fewer than 10 samples (e.g., PXD014058, and PXD004452),

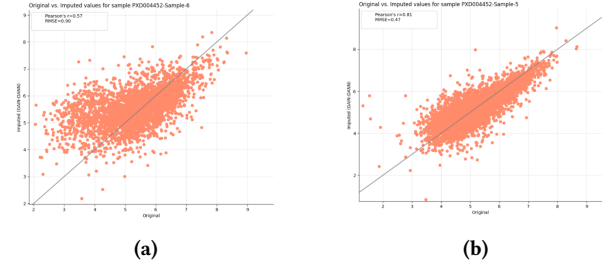


Fig. 4. Scatter plots showing the correlation between observed and imputed protein intensities for two samples from project PXD004452 of the HeLa-Reduced dataset (a) sample with the lowest performance, (b) sample with the highest performance.

exhibited substantially weaker performance, with RMSE values ranging between 0.76 and 0.858 and Pearson correlations lower than 0.6. This possibly reflects that having a limited sample representation hinders the model ability to capture project-specific characteristics important for the learning task: imputation.

Nevertheless, the sample size alone does not fully explain the variability. For instance, project PXD000296 with only 4 samples achieved a Pearson correlation of 0.907, outperforming other small sample projects with smaller missingness, namely PXD014058 with 4 samples and an overall missingness of 8.1% obtained only a Pearson of 0.513 and an RMSE of 0.858.

The aggregated values of the Pearson correlation and RMSE provide a global assessment of the model performance across the projects. While scatter plots of the original values versus the predicted values, enable a more granular sensitivity of the imputation quality at the sample level. Figure 4(a) illustrates the relationship between the original and imputed protein intensities for the sample with lowest performance within the PXD004452. In this sample, the correlation between the original sample and the imputed sample is 0.57 and the RMSE is 0.90, aligned with the relatively lower performance of this project.

The distribution of points around the diagonal reveals that the imputed values generally follow the true intensities, but deviations are more pronounced at higher intensity ranges, suggesting a tendency towards underestimation. Following the same trend, the scatter plots from projects with higher performance consistently display a tighter clustering around the diagonal, reflecting more accurate imputations (Fig. 4(b)).

Overall, the GAIN-DANN model demonstrated strong imputation performance on the HeLa-Reduced dataset, with an average Pearson correlation of 0.83 ± 0.18 and an average RMSE of 0.42 ± 0.32 , taking 8 hours to train. Its performance is comparable to the performance of the baseline ProtoGain model, which achieved an RMSE of 0.73 and required approximately 8 minutes to train. However, it is important to note that the evaluation of the GAIN-DANN model is more rigorous than the validation of the ProtoGain. At the time of its development, reduced data availability constrained ProtoGain to be trained on all the data availability while at the same time being evaluated on a portion of that data, thus this can lead to

overfit. In contrast, the GAIN-DANN model was assessed using a cross-validation strategy.

HeLa-Reduced-NoPXD042233

Since the project PXD042233 accounts for the majority of the samples in the HeLa-Reduced, we wanted to inspect whether it enforces more pressure on the model to distinguish it from the other projects, potentially influencing the learned latent representation.

In reality, that does not seem to happen. The UMAP projection of the latent space seems to aggregate the samples in five main clusters. Interestingly, the projects with a more distinct distribution of protein intensities (PXD048325 and PXD054015), form their own cluster. This confirms the earlier suspicion that these datasets are indeed different from the remaining projects, possibly reflecting differences in experimental settings.

Additionally, projects with lower missingness, such as the project PXD013658, PXD014058 and PXD000296, are also clustered together in the lowest right side of the UMAP. The grouping aligns with their coverage similarity ranging from 0.45 to 0.55.

This led us to wonder if the latent representation captures protein completeness instead of protein abundance. However, the Spearman correlation between the pairwise coverage similarity and the latent space distance between the centroids of the projects, was only -0.267 suggesting that there is only a moderate inverse relation. In contrast, the correlation between abundance similarity and latent distance between the centroids was -0.058 suggesting that the clustering in the latent space is not only explained by the abundance distribution. As such, the encoder appears to capture other sources of variation beyond protein coverage and abundance, it may possibly reflect biological variation or different experimental settings.

Overall, the model achieved similar performance to the HeLa-Reduced setting, suggesting that removing PXD042233 did not impact the imputation quality neither the model's generalization.

3.2.2 Cross-Dataset. To assess whether pretrained representations can improve the imputation of missing values, we conducted a cross-dataset evaluation using the transfer learning strategies described in Section 2.6.

We pretrained models on datasets of the same tissue type, leaving the dataset with the highest missingness within that tissue type as the target. This design reflects the intuition that small and highly incomplete datasets should benefit the most from transfer knowledge compared to models trained solely on that limited target data. In addition, most of the source datasets exhibited very low levels of missingness, in some cases close to zero, which makes them ideal candidates for knowledge transfer. To better understand whether the knowledge transfer indeed helped imputation quality, a baseline ProtoGain model was also trained directly on the target dataset. If the retrained model with the transfer knowledge outperformed the baseline, we can then argue that the transfer knowledge was indeed beneficial. Conversely, if the baseline model yielded better results, conclusions would need to be drawn more cautiously, since very small target datasets are prone to overfitting.

The source dataset for each tissue was constructed by merging all samples belonging to the same tissue type. Proteins absent from individual projects were set to *NaN* in the union dataset. To maximize performance, we restricted the union dataset to the intersection

of proteins shared with the target dataset. This ensured that the pretrained model would focus on imputing proteins relevant to the target dataset. In all cases, the domain of each sample corresponds to the PRIDE project it belonged to.

All models were pretrained on the source dataset for 200 epochs, with an early stopping patience of 10 epochs and the following hyperparameters: learning rate of 1×10^{-4} , two hidden layers for both the encoder and decoder, a dropout rate of 0.3, $\alpha = 2$, $\beta = 3$ and $\gamma = 1$. During fine-tuning, the same hyperparameters were used. In addition, the domain adaptation weight λ_p was fixed at 0.2 in all experiments. This value was chosen empirically as it provided the most stable results, ensuring that the adversarial contribution remained constant across training epochs and thus facilitated the adaptation to the target dataset. This aligns with the objectives of the fine-tuning stage that was designed to take advantage of the pretrained model to reduce computational time while improving imputation quality.

Brain tissue type

For the brain tissue, the source dataset was constructed by merging samples from projects PXD010154, PXD020192, PXD012131, and PXD005445, resulting in a dataset with 35 samples and 13,273 proteins. The project PXD000561 was chosen as the target dataset since it exhibited the highest missingness among the brain datasets. Restricting both the source and the target dataset to their protein intersection yielded 4,034 common proteins, leading to an overall missingness of 6.79% for the source dataset and 35.96% for the target dataset. Pretraining on the source dataset required approximately 31 seconds.

The results are consistent with our expectations. For the first strategy (see Section 2.6.1), where the encoder is completely frozen and only the decoder and GAIN module are retrained, the model achieved an RMSE of 1.489 after 60 epochs. Allowing partial unfreezing of the encoder and decoder yielded an RMSE of 1.202 after 100 epochs (see Section 2.6.1). The higher number of epochs in the second strategy is motivated by the higher number of trainable parameters.

Finally, the domain-adversarial adaptation strategy provided the strongest results (see Section 2.6.3). In this configuration, the model achieved an RMSE of 0.631 on the target dataset within only 20 seconds, whereas training ProtoGain directly on the target dataset required 1 minute and 18 seconds and achieved a higher RMSE of 0.946. These results support the hypothesis that small datasets with high missingness can particularly benefit from pretrained representations.

4 Discussion

The proposed GAIN-DANN model integrates generative imputation and domain-adversarial learning to address high missingness and domain shifts in proteomics data. The results across both intra-dataset and cross-dataset experiments confirm that the adversarial training of the proposed model enables the model to extract domain-invariant representations, thereby improving imputation quality. These findings support the hypothesis that shared information across related domains and datasets can enhance imputation performance.

4.1 Intra-dataset

The intra-dataset setting evaluates the model ability to generalize across related domains that may suffer from distributional shifts while still sharing underlying biological information. This constitutes the foundation of this scenario: if related biological information exists, the model should be able to learn domain-invariant representations of the data that can generalize beyond domain-specific biases.

In this setting, the model was evaluated on a HeLa dataset, where all the samples share the same biological source. Under such conditions, substantial biological similarity among samples would be expected. However, the UMAP projection of the HeLa-Reduced dataset revealed clear clustering by project, indicating the presence of batch effects (Fig. 2(a)). While the impact of such effects are difficult to quantify in a missing-value context, they can introduce biases that negatively compromise downstream analysis.

The GAIN-DANN model successfully reduced these batch effects while achieving a lower RMSE than the baseline ProtoGain. This improvement suggests that the domain-adversarial component aided the learning of a more faithful representation of the data rather than technical noise.

The results demonstrates that even when the samples share the same biological source, domain shifts introduced by experimental conditions can still bias the imputation. Addressing domain shifts is therefore relevant not only for cross-dataset but also within datasets with the same biological source where technical variation may be present.

4.2 Cross-dataset

In the cross-dataset setting, the benefits are even more pronounced and relevant as this scenario corresponds to the primary motivation for the proposed model. The results clearly demonstrated that leveraging cross-dataset information can enhance imputation quality, particularly in data-limited conditions.

Among all the fine-tuning strategies, the domain-adversarial approach achieved the best overall performance. Its superior performance can be attributed to its emphasis in minimizing discrepancies between the source and target domains during fine-tuning. By enforcing domain-invariant representations in the latent space, the encoder was able to generalize across datasets, improving the imputation quality in the target domain.

The pretrained model not only converged faster than the baseline model, but it also achieved a substantially lower RMSE. This contrast underscores the impact of transferring pretrained representations rather than learning completely from scratch in data-limited settings. In addition to the small size of the target datasets, they also suffer from a substantial missingness, which consequently, reduces even further the data available to the model, hindering the model ability to learn robust information. In such cases, transfer learning proved to be particularly beneficial, allowing the model to leverage the structure and variability captured from the more complete source datasets to guide the imputation process.

The alternatives fine-tuning strategies performed worse than the baseline model. This indicates that mere retraining of the last layers of the encoder and/or decoder was insufficient to overcome the

domain shift, and that a more in-depth adaptation is needed by full retraining. In other words, an adaptation of the latent space through adversarial alignment allowed the model to adapt to the target distribution while retaining generalizable representations.

These findings validate the hypothesis that pretrained representations can substantially improve imputation performance in small and incomplete proteomics datasets. They also suggest that pretraining on larger datasets can further expose the model to relationships between proteins that may not be present or detectable in smaller target datasets. Once learned, such representations can then be used to aid imputation in data-limited conditions.

4.3 Computation considerations

While the GAIN-DANN model consistently outperformed ProtoGain in terms of RMSE in both the intra-dataset and the cross-dataset scenario, this improvement came at a computational cost. For instance, in the HeLa-Reduced dataset, ProtoGain achieved an RMSE of 0.73 in approximately 8 minutes, whereas the GAIN-DANN model achieved an RMSE of 0.42 taking 8 hours. This difference was, however, expected. Since the GAIN-DANN architecture incorporates the GAIN module alongside three other modules, which increase the number of trainable parameters. Furthermore, the adversarial interplay between the modules introduces even more competing objectives, which naturally have slower convergence. This difference will only be aggravated the larger the dataset, which is a common scenario in proteomics.

Nevertheless, the cross-dataset experiments demonstrated that once pretrained, the GAIN-DANN model reached convergence substantially faster than the baseline model, achieving an RMSE of 0.63 compared to 0.95 while reducing the computational time by approximately 75%. This highlights the advantage of incorporating pretrained representations that can easily adapt to new datasets with minimal additional training.

5 Conclusions

In this work, we addressed the problem of missing value imputation in proteomics data, a critical step to ensure reliability in the downstream analysis. Building upon previous work in our group, that adapted GAIN to proteomics, we proposed GAIN-DANN, a hybrid model that integrates Generative Adversarial Imputation Networks with Domain-Adversarial Neural Networks.

The model was designed to exploit shared information across related domains and datasets by learning domain-invariant representations capable of generalizing to new conditions. Since working with multiple domains or datasets inevitably introduces domain shifts, the model was developed to explicitly address these shifts through its adversarial learning strategy. Across experiments, GAIN-DANN effectively mitigated these shifts improving the model generalization capacity.

In intra-dataset evaluations on the HeLa dataset, GAIN-DANN outperformed the baseline ProtoGain, achieving lower reconstruction error while reducing batch effects. In cross-dataset scenarios, the improvements were even more pronounced. The GAIN-DANN model converged faster and achieved a lower RMSE than the ProtoGain model trained from scratch. This demonstrated that previously

learned representations can be reused to enhance imputation in smaller and noisier datasets.

In summary, this work introduced the GAIN-DANN framework for missing data imputation in proteomics. Through its adversarial strategy, the model is able to learn domain-invariant representations, reduce batch effects and improve imputation quality across both intra and cross-dataset settings. These findings demonstrate that incorporating domain adaptation and generative imputation models can indeed enhance the imputation process.

5.1 Limitations and Future Work

While the proposed GAIN-DANN model consistently outperformed the baseline ProtoGain in terms of imputation quality, it also required longer training times. This can be viewed as a limitation, since the addition of more modules combined with the adversarial optimization dynamics can inherently slow convergence. However, it is important to note that the training time seems to be largely influenced by the size of the dataset. For instance, training on the HeLa-Reduced dataset (4,820 samples and 9,013 proteins) required approximately 10 hours, while training on HeLa-Reduced-NoPXD042233 (562 samples and 9,013 proteins) require approximately 75 minutes. In contrast, training on the brain source dataset (35 samples and 4,035 proteins) only required 31 seconds. This disparity highlights the scalability problem common when dealing with large-scale proteomics datasets. Nevertheless, the adversarial training strategy not only improved the imputation quality but also helped mitigate batch effects, which are known to compromise downstream analysis.

Moreover, once trained the learned representations can be efficiently reused for imputations in other datasets, substantially reducing the training time while achieving better results compared to training from scratch. Thus, although the initial training phase is more demanding, the generalization capability of GAIN-DANN makes it a strong candidate for imputation in smaller more incomplete proteomics datasets.

Future work could explore incorporating protein-protein interactions (PPI) to improve protein abundance estimation. From a graph-theoretic point of view, this can be formulated as a link prediction problem, where the objective is to predict the likelihood of a missing or unobserved edge between nodes in the network. Integrating Graph Neural Networks could further enhance imputation by capturing both node-level (protein abundance) and edge-level (interaction-based) dependencies between proteins. Potential directions include building GNNs directly from the proteomics data, build a GNN from the proteomics data considering the PPI interactions where the proteins are involved or employing adversarially trained GNNs inspired by GrapAT [2].

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